

Mengdi JIA

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👤 <https://mengdijia.github.io/>

EDUCATION

Anhui Agricultural University

Master of Engineering in Agricultural Engineering

2020.09 - 2023.06

- **GPA:** 3.54 / 4.0 (Top 10%)

College of Modern Science And Technology Hebei Agricultural University

Bachelor of Engineering in Mechanical Design, Manufacturing & Automation

2014.09 - 2018.06

- **GPA:** 3.52 / 4.0 (Top 3%)
- **Honors:** First-Class scholarship (annually); **Outstanding** Graduation Project; **Champion**, World Robot Olympiad (China)

RESEARCH EXPERIENCES

OmniSpatial: Towards Comprehensive Spatial Reasoning Benchmark for Vision-Language Models

Mengdi Jia^{1*}, Zekun Qi^{14*}, Shaochen Zhang², Wenyao Zhang³⁴, Xinqiang Yu⁴, Jiawei He⁴, He Wang⁴⁵, Li Yi^{16†} ICLR, 2026

- <https://arxiv.org/abs/2506.03135>
- **Benchmark:** Proposed **OmniSpatial**, a novel and comprehensive spatial reasoning benchmark addressing limitations of existing vision-language evaluations predominantly focused on high level spatial tasks.
- **Categorization Framework:** Established a structured categorization comprising four dimensions—dynamic reasoning, complex spatial logic, spatial interaction, and perspective-taking—to enhance evaluation complexity and breadth.
- **Dataset Construction:** Constructed the OmniSpatial dataset by crawling and curating diverse visual data from global sources, spanning various scenes, resolutions, illumination conditions, and weather scenarios, ensuring realistic and comprehensive evaluation contexts.
- **Model Evaluation and Insights:** Performed thorough evaluations of state-of-the-art vision-language models (e.g., ChatGPT O3, Gemini-2.5-Pro), identifying notable deficiencies in advanced spatial reasoning and providing actionable insights for future research.
- **Reasoning Enhancement:** Enhanced spatial reasoning capabilities of VLMs through integrating auxiliary models using a chain-of-thought approach, demonstrating effective strategies for complex multimodal reasoning.

Experimental Investigation on the Crack Propagation Principle of Pecan under Heating State

Mengdi Jia¹ Master's Thesis, 2023

- DOI: 10.26919/d.cnki.gannu.2023.000742
- Developed a real-time weight and temperature monitoring system using LabVIEW
- Implemented a crack detection algorithm with YOLOv8, and constructed a moisture prediction model using near-infrared spectroscopy and BP neural network.
- Constructed a moisture prediction model using near-infrared spectroscopy and BP neural network.

Biophotonics Lab, Dept. of Electronics, Tsinghua University

Intern

2019/12 - 2020/09

- Developed biomedical device components using OpenCV, C++, Qt and SolidWorks for photoacoustic imaging systems.

SKILLS

English Proficiency: Fluent

Programming & Tools: Python, C++, C, MATLAB, LaTeX, PyTorch, NumPy, Pandas, OpenCV, Linux, Jetson, STM32

WORK EXPERIENCES

Noetix Robotics Co., LTD

Mechatronics Engineer (Structural Lead)

2025/07 - Present

- To C Scenario - Small Bipedal Robot for Companionship:
 - a. Optimized dual-leg folding scheme by designing limb positions and leg folding angles.
 - b. Calculated and evaluated torque and speed demands of each joint module under high-dynamic load conditions to support customized joint module development.
 - c. Served as a Project Manager, developed product definition and product requirement documents, ensured information consistency among technical team members, and led technical solution analysis.

- To B Scenario - Large Bipedal Robot for Inspection:
 - a. Optimized shaft system design to address intermittent high-intensity impacts on knee joints; replaced graphite copper sleeves with PEEK.
 - b. Proposed a novel bidirectional symmetric folding knee joint structure, resolving parallelogram dead-point issues and improving linkage force transmission efficiency.
 - c. Designed foot-end structures for the robotic foot.
- Humanoid / Character Robot:
 - a. Designed robot body structures and integrated sensing layouts for humanoid and character-style robots, including sensor placement and extrinsic calibration.
 - b. Exported URDF models and performed simulation-based training and validation.

Beijing Donghong Zhiyuan Medical Technology Co., LTD

Mechatronics Engineer (Project Leader)

2024/05 - 2025/06

- Leading electromechanical design and lifecycle management of surgical instruments and endoscopes.

Beijing Precision Medical Technology Co., LTD

Project Engineer

2023/07 - 2024/04

- Developed robotic end-effectors and calibration methods for MR-guided surgical systems.

Solidreamer Co., LTD

Co-founder

2014/12-2017/06

- Provided robotics and STEM training for teenagers, organized workshops, designed educational programs.

PUBLICATIONS AND PATENTS

OmniSpatial: Towards Comprehensive Spatial Reasoning Benchmark for Vision-Language Models

Mengdi Jia^{1*}, **Zekun Qi**^{14*}, **Shaochen Zhang**², **Wenyao Zhang**³⁴, **Xinqiang Yu**⁴, **Jiawei He**⁴, **He Wang**⁴⁵, **Li Yi**^{16†} *ICLR, 2026*

- <https://arxiv.org/abs/2506.03135>

Experimental Research on Cotton Seed Depth Detection System Based on Magnetic Field

Jia Kang, **Nan Wang**, **Haiyong Jiang**, **Pengyun Xu**, **Mengdi Jia**, **Limin Shao**

Graduation Project, Mar. 2021

- Published in *Journal of Agricultural University of Hebei*, Vol. 44 No. 2.
- DOI:10.13320/j.cnki.jauh.2021.0033.
- Built a parallel-arm robot capable of detecting seed depth in soil while driving through farmland.

A device and method for light emission protection of photoacoustic probes based on transparent capacitive films

Wu Zhen, **Wang Xiaojun**, **Song Hongfei**, **Jia Mengdi**, **Fang Chenyu**

Nov. 2023

- Patent No.: CN 113827183 B

Monitoring devices and methods for visible and invisible light energy of lasers

Song Hongfei, **Wang Xiaojun**, **Wu Zhen**, **Fang Chenyu**, **Jia Mengdi**, **Wei Shengyi**

Jan. 2022

- Patent No.: CN 113970371 A

A differential analog transmission system for the acquisition of photoacoustic signals

Song Hongfei, **Wang Xiaojun**, **Wu Zhen**, **Fang Chenyu**, **Jia Mengdi**, **Wei Shengyi**

Jan. 2022

- Patent No.: CN 113966996 A

A laser emission protection method applicable to photoacoustic imaging systems

Wu Zhen, **Wang Xiaojun**, **Song Hongfei**, **Jia Mengdi**, **Fang Chenyu**

Dec. 2021

- Patent No.: CN 113827184 A

A differential analog transmission device for photoacoustic signal acquisition

Song Hongfei, **Wang Xiaojun**, **Wu Zhen**, **Fang Chenyu**, **Jia Mengdi**, **Wei Shengyi**

Jan. 2021

- Patent No.: CN 212326383 U

Monitoring devices for visible and invisible light energy of lasers

Song Hongfei, **Wang Xiaojun**, **Wu Zhen**, **Fang Chenyu**, **Jia Mengdi**, **Wei Shengyi**

Jan. 2021

- Patent No.: CN 212340426 U

A seed depth detection system based on magnetic field

Jia Mengdi, **Feng Yongfei**, **Jiang Haiyong**, **Zhou Yongjie**, **Wang Nan**

Apr. 2019

- Patent No.: CN 208736340 U

An Extrusion Device for Additive Manufacturing of Flexible Materials Using Hard Materials

Jia Mengdi

Apr. 2018

- Patent No.: CN 207273878 U

A Peach Flower Stamen Cutting Mechanism

Du Yujie, **Jia Mengdi**

Mar. 2018

- Patent No.: CN 207054379 U